

Synergy-Based Prosthetic Control

Decoding a handful of noisy muscle signals into a dexterous, 24-joint prosthetic hand

Skandalaris Fall Hackathon · Team Synergy
Repo: github.com/guojohn1/prosthetic-synergy-control
Regenerate all numbers: `./run_real.sh`

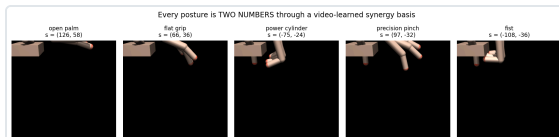
The one-sentence idea: a hand has about 24 independently-movable joints, but an amputee's muscles only give a prosthetic controller 4–12 noisy signals — nowhere near enough to drive each joint directly. Real hand postures barely use 24 independent dimensions anyway: they mostly live in a space of a couple of “synergies,” coordinated muscle groupings the brain already uses. So instead of muscle signal → joint, we map muscle signal → **2 synergy numbers** → a fixed basis (learned once, from watching intact hands) that expands those 2 numbers into all 24 joints. We built an environment that measures whether this survives real amputee data, and an interface that shows the answer in seconds.

The problem, in plain terms

Today's myoelectric prosthetic hands mostly cope with the “few signals, many joints” mismatch by **mode-switching**: the user flexes in a set pattern to cycle through a menu of preset grips (pinch, power grip, point) and picks one. It's slow and unnatural — a big reason many devices get abandoned. And it gets worse for exactly the people who need it most: the more forearm that's gone, the fewer independent muscle signals are left to read.

The idea: fix the coordinate system, not the hardware

The hand's 24 joints don't move independently — making a fist flexes dozens of joints together in a predictable pattern. So posture compresses into a handful of numbers (“synergies”) without losing much information. We learn that compression once, from recordings of *intact* hands grasping everyday objects, and freeze it. From then on, controlling the hand means producing just 2 numbers: a small, quickly-calibrated personal decoder turns muscle signal into those 2 numbers, and the frozen basis expands them into a full, coordinated 24-joint posture.

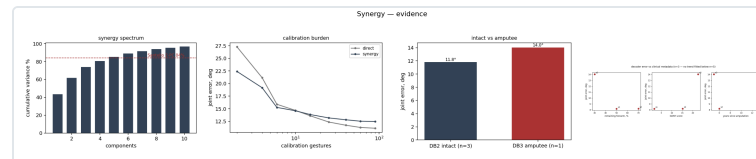


Five grasps, each produced from just two numbers through the same fixed synergy basis — simulated on a 24-DOF hand model (MuJoCo).

What we actually tested

We ran the full pipeline (synergy basis → EMG decoder → joint reconstruction) on **real recordings** from NinaPro: intact subjects (DB2) and transradial amputees (DB3), real forearm EMG, real glove ground truth — every number at right is measured, not assumed.

The evidence



Left to right: (1) how much of real hand-posture variation is captured by N synergies; (2) decoder accuracy vs. how many calibration gestures a new user provides; (3) accuracy on intact vs. amputee subjects; (4) the same accuracy plotted against each amputee's clinical profile.

61.6%

VARIANCE, 2
SYNERGIES (VS. ~84%
CITED)

+17.9%

SYNERGY ADVANTAGE
AT 2 GESTURES

11.8° → 14.0°

INTACT → AMPUTEE
ERROR

1. Is hand posture really low-dimensional? — yes, but less than the textbook says

A classic 1998 study (Santello et al.) found 2 synergies explain ~84% of real hand-posture variation. On our real CyberGlove data, 2 synergies explain **61.6%** (3 synergies: 73.9%) — we report the weaker number, because the architecture has to survive real data, not just the citation.

2. Does the synergy shortcut actually help? — only when the user has given almost nothing

Two decoders on identical data: “direct” learns each of 24 joints separately; “synergy” learns just 2 numbers, then reuses the frozen basis. At 2 calibration gestures, synergy is **17.9% more accurate**; the two cross over around 10 gestures; by 90 gestures, direct is **12% better**, since 2 numbers are a hard ceiling direct fitting doesn't share. Backwards from a hype curve — and exactly right for an amputee who can only give a handful of practice reps at setup.

3. Does it hold up on amputee data, not just intact hands? — yes, with a measurable gap

The identical pipeline on an amputee subject (usable glove ground truth) gives **14.0°** average joint error vs. **11.8°** on intact subjects — a real but modest **2.2°** gap, on **n=1**: two of three amputee subjects have no hand left to wear a glove, so we scored them against intact-hand prototypes instead and kept them out of this gap number, rather than mixing two kinds of “truth.”

What's built and working today

- A MuJoCo simulation of a 24-DOF hand, driven live by 2 numbers
- A synergy basis fit to real intact-hand posture data (NinaPro DB2)
- A ridge decoder: real forearm EMG → synergy coordinates → joint angles
- The identical pipeline re-run, unmodified, on real amputee data (DB3)
- A ghost-comparison UI: glove-intended posture beside the posture reconstructed from muscle signal alone, scrubbable, with a live per-frame error readout
- A training panel where each practice rep moves a live projection along the calibration curve above — practice doubles as setup, no separate ritual

Honest limitations

- Real data reproduces the low-dimensionality premise more weakly than the founding citation (61.6% vs. ~84%)
- The synergy advantage is real but temporary — inverts past ~10 calibration gestures
- Intact-vs-amputee gap rests on n=1 with usable glove ground truth; clinical-metadata pattern is n=3, unfitted
- No clinical claims — a control architecture and measurement environment, not a validated medical device

What's next

The frozen basis is both floor and ceiling: why the system needs almost no setup, and why it caps out with more data. The intact-amputee gap proxies the fine synergies amputation destroys. Next: a **subject-adaptive refinement** to reconstruct those per user, paired with a **ground-truth rig** amputees can actually wear — two of three subjects couldn't use the standard glove at all.